

battery pack when purchasing a laptop.

Barring the odd one here and there, most of the LCD screens on laptops are usable. Some are exceptional, such as the specially-coated screens offered by Sony or Sharp (with exceptional price tags too); some are not very usable under certain lighting conditions, such as the 'mirrored' Toshiba Satellite. But, on the whole, they are all usable for the task at hand, which is mostly reading, writing and some portable cinema, if you are lucky with time and if your battery allows.

Before purchasing a laptop, ensure that you are comfortable with the viewing angle—place it in a position you will most frequently use: on your lap, on a table, in a cramped airplane seat, or seated in a local train. If the screen is readable at the angles you try out, you are good to go.

Another noteworthy aspect is that of the hard disk speed, but all the laptops we tested had 4,200 rpm disks, which is good

enough. Just remember that faster, though better, would also drain batteries quicker.

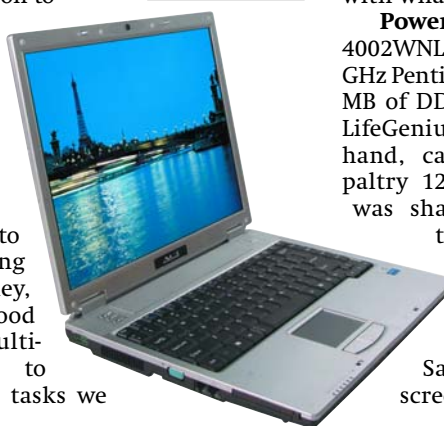
Lastly, on build quality—IBM laptops are the best in this regard, bar none. The quality of their keyboards is on par with, if not above, that of desktop keyboards, their machines are build extremely well, and to last. Time will tell whether the IBM-Lenovo deal will affect this; notably, Thinkpads will still be made under the aegis of IBM for at least one more year.

We will now move on to the crux of this section—the performance of the laptops we received, starting with the Workhorses.

THE WORKHORSES

Workhorses are built to take care of demanding computing needs. They, therefore, include good I/O, graphics and multimedia sub-systems to enable you to tackle tasks we

ACI Emerald C2



usually perform on our desktop PCs. We rounded 12 models from nine brands under this category. HP and Dell could not send in their models in time for our test, despite aggressive follow-ups from our end.

A workhorse is generally employed both at home (where you will likely use it for work) and at work (if the boss is looking, for work). These machines, therefore, are powerful enough to pull the requisite load. Let's look at their features, starting with what makes them tick.

Power: The Acer TravelMate 4002WNLCi comes with a fast 1.6 GHz Pentium M processor and 512 MB of DDR333 RAM. The Wipro LifeGenius 1000D, on the other hand, came equipped with a paltry 128 MB of RAM, which was shamelessly shared with the graphics chip—its performance did suffer on account of this. The Acer, Zenith Strategist and Toshiba Satellite all had wide screens, which are a plus

How We Tested

The laptops were assessed on four main parameters—features, performance, usability and price. The weightages applied to these parameters varied from one category to another depending on the relevance of a particular parameter to that category.

We used a fresh copy of Windows XP Professional SP1 as the OS. After the installation, all the latest drivers were loaded for optimum performance.

Features

We noted features such as the type of RAM, hard disk capacity, type of optical drive, types of ports, connectivity options, and so on.

Usability

A laptop's usability is evaluated on the basis of how simple or difficult it is to handle in day-to-day life—primary importance was given to weight, dimensions, and other ergonomically-inclined issues such as keyboard feel, touchpad sensitivity, etc.

Package contents

We noted whether the vendor provided an OS, a recovery CD, a user's manual and driver CDs with the laptop. We also noted the extra software provided, and the accessories—power adapter, carry case, etc.—bundled along.

Performance

To gauge the performance of each laptop, we ran a battery of tests to evaluate each sub-system. The following benchmarking suites were used to test the laptops.

PCMark 2004: This is a system-wide benchmark that tests individual components such as the processor, memory, hard drive and the graphics sub-system. It returns aggregated scores as well as individual scores by running different applications used in day-to-day work, such as file encryption, virus scanning and so on.

SiSoft Sandra 2005 Professional SR1: This was used to evaluate CPU, memory and hard disk performance.

Ziff Davis Business Winstone 2004: ZD Bench's Business Winstone—a benchmark that tests the system with applications that are used on daily basis—evaluates complete system performance, and returns a unified score.

Video Encoding: A 100 MB VOB file was encoded using Dr. Divx 1.0.6 to the AVI format, and the time taken to encode the file was noted.

Gaming benchmarks: *Call of Duty* was used because it doesn't work on graphics chipsets that don't support Hardware T&L (Transformation and Lighting), and tests the OpenGL

graphics subsystem. *HalfLife 2* supports different versions of the DirectX API. The games were run at 800 x 600 and 1024 x 768, and the average frame-rate was noted.

Screen display tests: DisplayMate Video Edition was used to evaluate the quality of the LCD display. We used the test to gauge the sharpness and level shift of the LCD screens. Pixel Persistence was tested using the PassMark Monitor Test suite, in which white moving blocks were observed for the tearing effect, and the screens were rated accordingly. The laptops were then taken outdoors to see how viewable the text and video on the screen was in sunlight. The laptops were placed such that the sun fell on the screen at a slant, rather than perpendicular to the screen.

Digit Battery Meter: In order to stress the battery to the utmost, we ran a VOB file until the battery ran out, to gauge whether the notebook battery would last a full movie—pretty real-world.

Wi-Fi Test: To evaluate data throughput over Wi-Fi, we used the NetCPS program—a utility that pumps in data to check the TCP/IP connection. We also copied 52 MB of assorted data, and streamed a movie file, to further check Wi-Fi performance.